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CS 2614 || Homework 5

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1-16. Design a combinational circuit with three inputs x, y, z and three outputs A, B, C. When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is one less than the input.

x	y	z	A	B	C
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	0	1	1
1	0	1	1	0	0
1	1	0	1	0	1
1	1	1	1	1	0

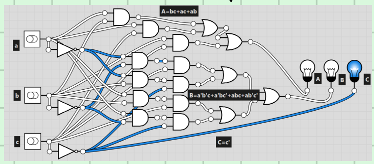
0 → 1
1 → 2
2 → 3
3 → 4
4 → 3
5 → 4
6 → 5
7 → 6

$A = yz + xz + xy$

$B = x'y'z + x'yz' + xyz + xy'z'$

$C = z'$

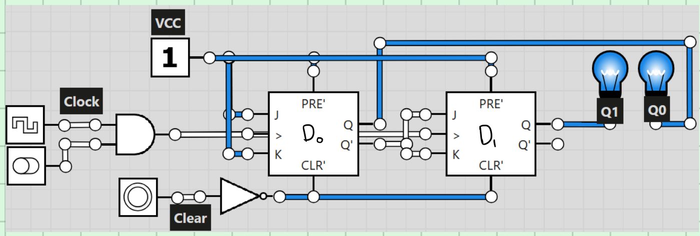
Preferred the correct variables are being used.
x → a
y → b
z → c



1-20. Design a 2-bit count-down counter. This is a sequential circuit with two flip-flops and one input x. When $x = 0$, the state of the flip-flops does not change. When $x = 1$, the state sequence is 11, 10, 01, 00, 11, and repeat.

$D_1 = x'Q_1 + x(Q_1 \odot Q_0)$

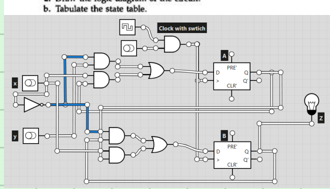
$D_0 = Q_0 \oplus x$



1-19. A sequential circuit has two D flip-flops A and B, two inputs x and y, and one output z. The flip-flop input equations and the circuit output are as follows:

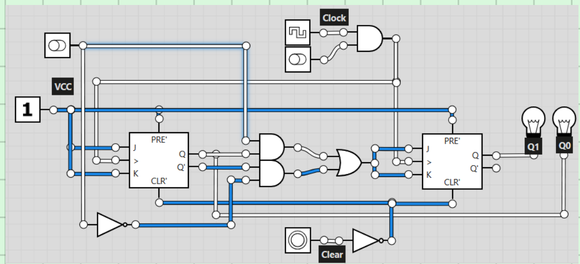
$$D_A = x'y + xA$$
$$D_B = x'B + xA$$
$$z = B$$

a. Draw the logic diagram of the circuit.
b. Tabulate the state table.



Current State	Inputs	Next State	Output
A B	x y	A_{next} B_{next}	z
0 0	0 0	0 0	0
0 0	0 1	1 0	0
0 0	1 0	0 0	0
0 0	1 1	0 0	0
0 1	0 0	0 1	1
0 1	0 1	1 1	1
0 1	1 0	0 0	1
0 1	1 1	0 0	1
1 0	0 0	0 0	0
1 0	0 1	1 0	0
1 0	1 0	1 1	0
1 0	1 1	1 1	0
1 1	0 0	0 1	1
1 1	0 1	1 1	1
1 1	1 0	1 1	1
1 1	1 1	1 1	1

1-21. Design a sequential circuit with two JK flip-flops A and B and two inputs E and x. If $E = 0$, the circuit remains in the same state regardless of the value of x. When $E = 1$ and $x = 1$, the circuit goes through the state transitions from 00 to 01 to 10 to 11 back to 00, and repeat. When $E = 1$ and $x = 0$, the circuit goes through the state transitions from 00 to 11 to 10 to 01 back to 00, and repeat.



5. Safe is moved & switch closed OR closet opened & after hours OR closet open & control switch open

B' (safe not in normal position), A, D' (closet not open), C' (not within work hours), D' (not closed), A' (not closed)

⇒ $Alarm = (B'A) + (D'C') + (D'A')$

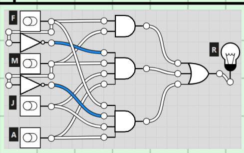
6. F = Father, M = Mother, J = John, A = June, R = Restaurant

Agree = (FM) + (F'M')
If they agree R (restaurant decided) is = F or M since they would be the same

Majority$_H$ = (every combination were 3 or more true)

majority wins except parents agree
tie result in chicken

If agree is true, R = F, otherwise, R = majority$_H$
if tie & agree' default is chicken



R = (Agree · F) + (Agree' Majority)

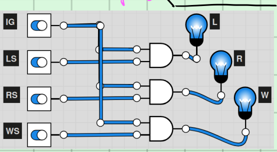
Agree · F = ((F·M) + (F'·M'))·F = FM

Agree' = ((FM) + (F'M'))'
= (FM)' · (F'M')'
= (F'+M') · (F+M)
= F'F + F'M + M'F + M'M
= 0 + F'M + M'F + 0
= F'M + M'F

R = (F·M) + ((F'M + M'F) · Majority$_H$)

R = (FM) + (F'MJA) + (M'FJA)

7. Automotive subsystems



8. Count number of 1's

a)

A	B	C	Number of 1's	X_1	X_0
0	0	0	0	0	0
0	0	1	1	0	1
0	1	0	1	0	1
0	1	1	2	1	0
1	0	0	1	0	1
1	0	1	2	1	0
1	1	0	2	1	0
1	1	1	3	1	1

b) Simplify logic function

$X_1 = A'BC + AB'C + ABC' + ABC$

AB \ C	0	1
00	0	0
01	0	1
11	1	1
10	0	1

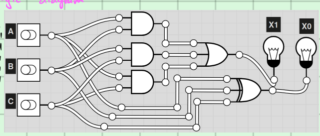
$X_1 = BC + AB + AC$

$X_0 = A'B'C + A'BC' + AB'C' + ABC$

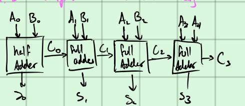
AB \ C	0	1
00	0	1
01	1	0
11	0	1
10	1	0

$X_0 = A \oplus B \oplus C$

c) Logic diagram



9. Block diagram of a 4bit adder using 1 half adder & 3 1bit full adders



10. Input 3 bit num, output 1 bit, out is 1 if in is divisible by 2 xor 3

a) Truth table

Num	A	B	C	Y
0	0	0	0	0
1	0	0	1	0
2	0	1	0	1
3	0	1	1	1
4	1	0	0	1
5	1	0	1	0
6	1	1	0	0
7	1	1	1	0

b) K-Map

A \ BC	00	01	11	10
0	0	0	1	1
1	1	0	0	0

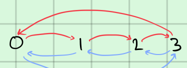
$Y = A'B + AB'C'$

c) Simplified boolean function for Y

$Y = A'B + AB'C'$

11) Design an Up/Down wraparound counter

a) State diagram (red: x=0, blue: x=1)



b) Truth table

Most significant bit

Q1 Q0	x	Q1*	J1	K1
0 0	0	0	0	X
0 0	1	1	1	X
0 1	0	1	1	X
0 1	1	0	X	1
1 0	0	1	1	X
1 0	1	0	X	1
1 1	0	0	X	1
1 1	1	1	1	X

Least significant bit

Q1 Q0	x	Q1*	J1	K1
0 0	0	0	0	X
0 0	1	1	1	X
0 1	0	1	1	X
0 1	1	0	X	1
1 0	0	1	1	X
1 0	1	0	X	1
1 1	0	0	X	1
1 1	1	1	1	X

same

current state | next state of Q' | Inputs of J/K

c) K maps

J_1:

Q_1Q_0 \ x	0	1
00	0	1
01	1	0
11	0	1
10	X	X

$J_1 = Q_0x + Q_1x'$

K_1:

Q_1Q_0 \ x	0	1
00	X	X
01	X	X
11	0	1
10	1	0

$J_1 = Q_0x' + Q_1'x$

J_0:

Q_1Q_0 \ x	0	1
00	1	1
01	X	X
11	X	X
10	1	1

$J_0 = Q_0'$

K_0:

Q_1Q_0 \ x	0	1
00	X	X
01	1	1
11	X	X
10	X	X

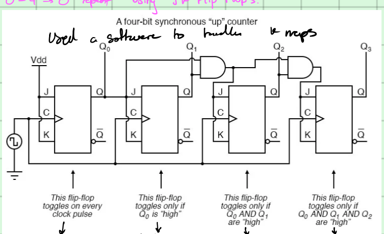
$K_0 = Q_0$

$J_1 = Q_0x + Q_1x'$
$K_1 = Q_0x' + Q_1'x$

$J_0 = Q_0'$
$K_0 = Q_0$

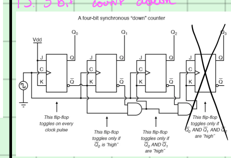
12. Design a binary counter that goes through the sequence 0 – 9 → 0 repeat using JK flip flops.

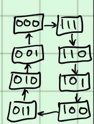
Used a software to handle k maps



$J_0 = 1$, $K_0 = 1$
$J_1 = Q_3'Q_1'Q_0$, $K_1 = Q_3'Q_1Q_0$
$J_2 = Q_3'Q_2'Q_1Q_0$, $K_2 = Q_3'Q_2Q_1Q_0$
$J_3 = Q_3'Q_2Q_1Q_0$, $K_3 = Q_3Q_2'Q_1'Q_0$

13. 3 bit count down





15. Create a sequential circuit with 2 D flip-flops

Current State (A B)	x	Next State (A' B')	D_A	D_B
0 0	0	0 0	0	0
0 0	1	0 1	0	1
0 1	0	0 1	0	1
0 1	1	1 1	1	1
1 1	0	1 1	1	1
1 1	1	1 0	1	0
1 0	0	1 0	1	0
1 0	1	0 0	0	0

K-maps:

D_A:

AB \ x	0	1
00	0	0
01	0	1
11	1	1
10	1	0

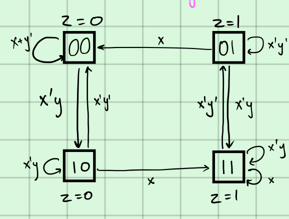
$D_A = Ax' + Bx$

D_B:

AB \ x	0	1
00	0	1
01	1	1
11	1	0
10	0	0

$D_B = Bx' + A'x$

14. a) State table for the sequential circuit

Present State	Inputs	Next State	Output
AB	xyxy	A(t+1)B(t+1)A(t+1)B(t+1)	z=Bz=B
0 0	0 0	0 0	0
0 0	0 1	1 0	0
0 0	1 0	0 0	0
0 0	1 1	0 0	0
0 1	0 0	0 1	1
0 1	0 1	1 1	1
0 1	1 0	0 0	1
0 1	1 1	0 0	1
1 0	0 0	0 0	0
1 0	0 1	1 0	0
1 0	1 0	1 1	0
1 0	1 1	1 1	0
1 1	0 0	0 1	1
1 1	0 1	1 1	1
1 1	1 0	1 1	1
1 1	1 1	1 1	1

b) State diagram